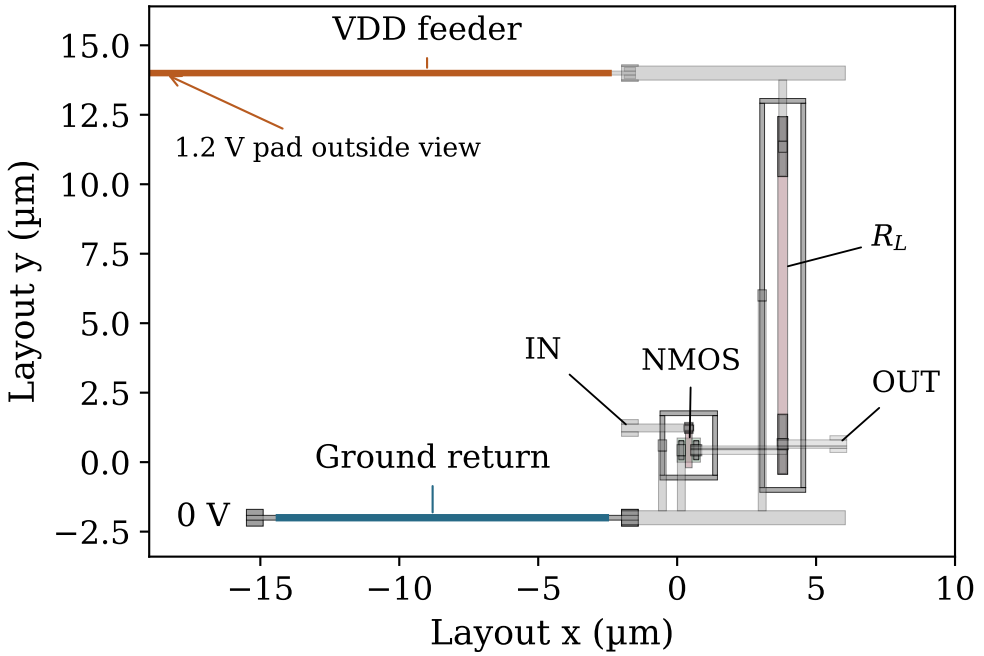
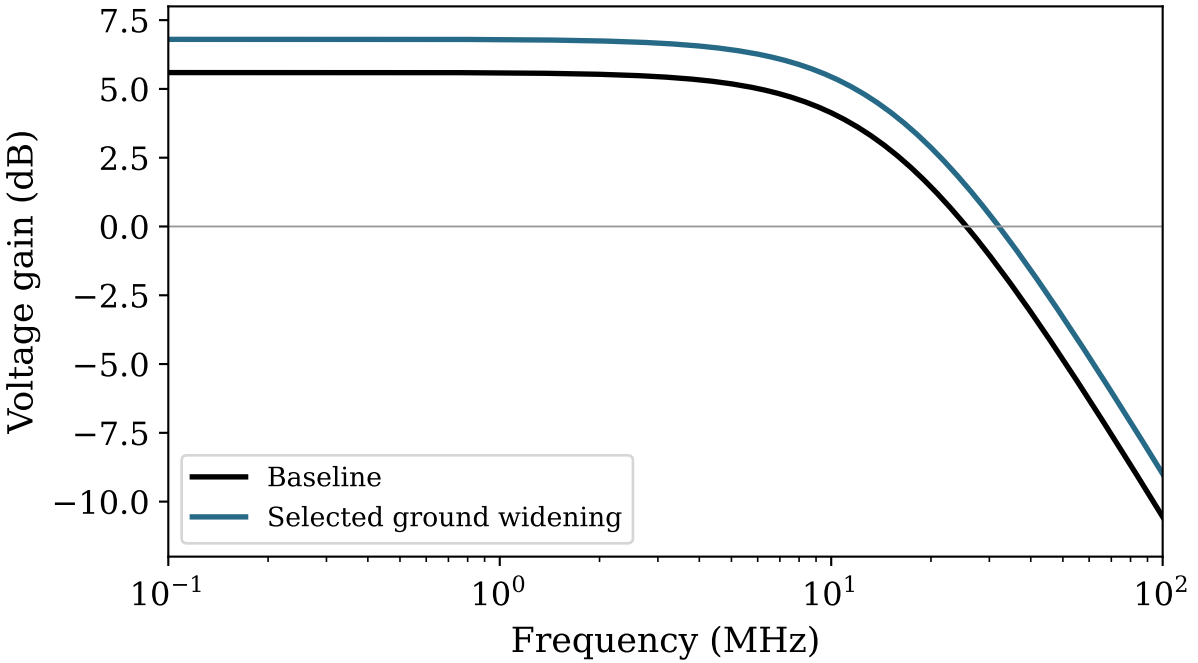
(a) Supply feeder and ground return



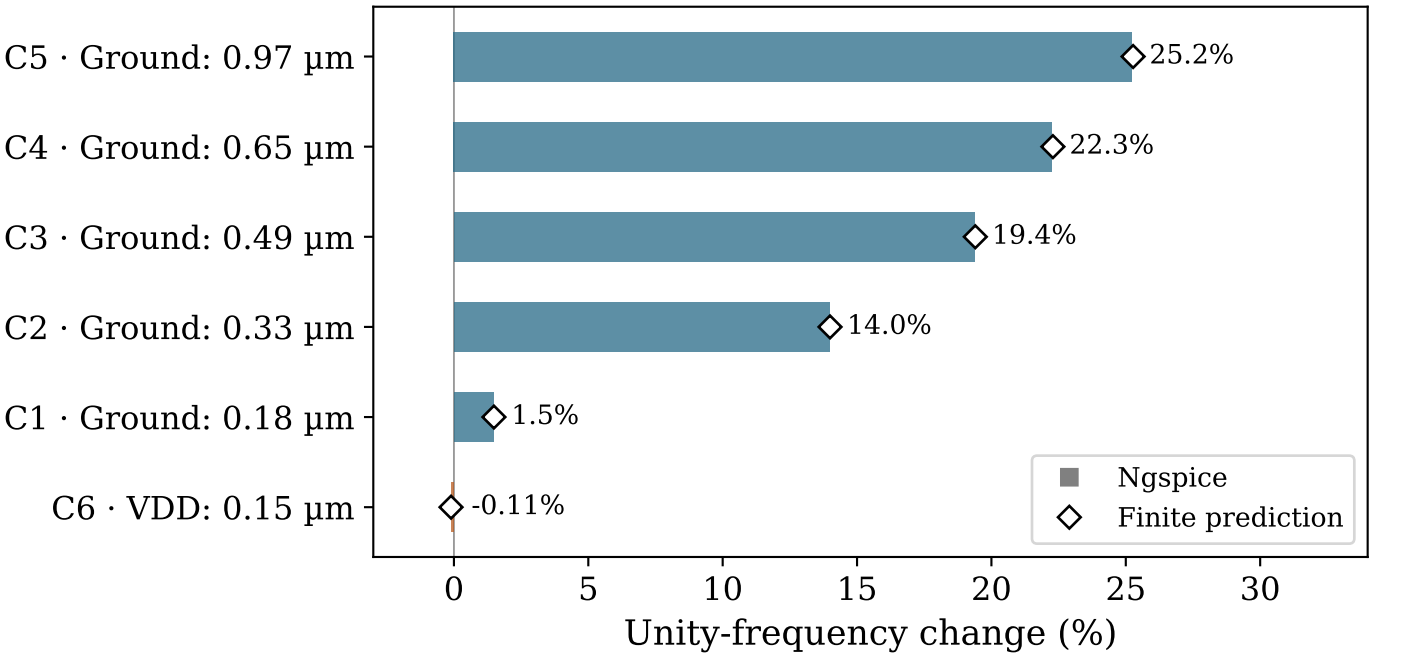
(b) Native ngspice response



(c) Measured operating conditions

Quantity	Baseline	Selected
Gain at 1 kHz (dB)	5.601	6.808
Unity frequency (MHz)	25.58	32.04
Supply current (µA)	45.86	52.72
NMOS source (mV)	51.65	21.54
Output bias (V)	0.689	0.613

1.2 V supply · 0.954725 V input · 1 pF load

Selected: 0.17 → 0.97 µm; added area 9.52 µm²

(d) Individual alternatives; budget 10 µm²